Simran Bilkhu

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Education

- **McMaster University** — *September 2024 – April 2026*
  M.Sc. Statistics (Thesis) — GPA 11.86/12 — Hamilton, Canada
  - Thesis [link]
    A Comparative Study of Bayesian Factor Models under PLT and UGLT Identification Constraints
- **University of Toronto** — *April 2024*
  H.B.Sc. Applied Statistics, Minor in Mathematics (Honours) — GPA 3.76/4.00 (final 2 yrs) — Toronto, Canada

Technical Skills

- **Languages & Tools:** R (tidyverse, terra, Stan, Shiny), Python (NumPy, Pandas, scikit-learn, Tensorflow), SAS, SQL, LaTeX, Git
- **Statistics & ML:** Bayesian inference, factor models, GLMs & GLMMs, survival analysis, time series, PCA, clustering (K-means, GMM, agglomerative), random forests, gradient boosting, neural networks, regularization methods (Lasso, Ridge, elastic net), Causal Inference (Structural Equation Models, Experimental Design, Instrumental Variables).
- **Platforms:** RStudio, Jupyter, Spyder, VS Code, Shiny (dashboards), Quarto, Github

Work Experience

- **McMaster University** — *Sep. 2024 – Dec. 2025*
  Teaching Assistant — (Statistical Methods, Linear Regression Using SAS, Intermediate Probability Theory) — Hamilton, ON
  - Conducted live coding sessions in **SAS**. Graded **data analysis** reports and provided feedback. Delivered tutorials and office hours for two undergraduate statistics courses; graded 500+ assessments and supported first-year students at the Math Help Centre 4 days/week.
- **Sodexo Canada** — *March 2020 – August 2022*
  Intern — Oakville, ON
  - Analyzed telecom contract and operational cost data to identify savings opportunities, contributing to over **$50,000** in reductions; built dashboards in Smartsheet to visualize project progress and present findings to leadership; managed structured data tracking across Canada-wide hardware deployments and asset management initiatives.

Projects

- **Diet Intervention and Gut Microbiome** — *January – April 2025*
  R, PCA, GMM, Random Forest — *Link*
  - Investigated how diet shapes gut microbial composition: preprocessed a 236×1,415-taxon matrix, reduced via PCA (20 components, >60% variance), clustered with GMM, and trained a Random Forest classifier (4,000 trees) achieving **91% accuracy** (adjusted Rand index 0.68).
- **Baseline Climate Partitions via Clustering** — *November 2024*
  Python, scikit-learn — *Link*
  - Defined stable baseline climate regions across North America to support ecological and policy analysis: benchmarked K-means, contiguity-constrained agglomerative (170 km), and PCA-then-agglomerative clustering on ∼**130K observations** across **19 bioclimatic features**. We found that there were 5 relevantly distinct climatological clusters in North America.
- **Bird Migration Count Forecasting** — *December 2024*
  R, Poisson Regression, Gaussian Random Fields — *Link*
  - Forecasted population trends of *Peregrine Falcons* and *Greater Roadrunners* using satellite climate data: modelled counts (13,642 obs, 2001–2005) from time, location, and weather features; a spatio-temporal mixed model was used to **reduce test error from 1,377 to 31** vs. ridge-penalised Poisson, with warmer temperatures positively associated with counts.
- **Agricultural Expansion & Wetland Change in Canada's Boreal Shield** — *September 2024*
  R, terra, Shiny — *Link*
  - Quantified a decade of agricultural expansion and wetland loss across Canada's Boreal Shield: processed 66 GeoTIFF land cover rasters (2013–2023) with `terra`, scaled estimates to km^2 per province, and deployed an interactive Shiny app for trend exploration.

Publications & Presentations

Paper: *Counter-Examples to Continuity of Measure in Uncountable Unions* (with N. M. Forman) [link].

Talks: Sparse Bayesian Factor Analysis [slides], Sparse PCA [slides], HMC [slides], Bayesian Workflow